INCH-POUND
MIL-PRF-28876/32
25 August 2023

PERFORMANCE SPECIFICATION SHEET

CONNECTORS, FIBER OPTIC, CIRCULAR, PLUG AND RECEPTACLE STYLE, MULTIPLE REMOVABLE TERMINI, SCREW THREADS, REPOSITIONABLE BACKSHELL, STRAIGHT, 45- AND 90-DEGREE, ENVIRONMENT RESISTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-28876.

REQUIREMENTS:

Design and construction. The internal surfaces of the repositionable backshell and accessories shall be free of any sharp edges in all configuration orientations (i.e., straight, 45-degree, and 90-degree) and shall provide a smooth, free-flowing pathway for the optical fibers transitioning though them. Internal volumes shall be large enough to accommodate the various cable types and termination methods expected to be employed with each shell size.

Repositional Backshell characteristics. Shall employ a repositionable body design.

a. The body shall have a design feature that allows for straight, 45-degree, and 90-degree orientations at a minimum.
b. The design shall maintain environmental sealing in all orientations.
c. The design shall provide a means of securely engaging and locking the various orientations in place to immobilize the configuration.
d. The manufacturer's backshell design shall have the same cable preparation length in the various orientations for each shell size.

Dimensions and spline configurations. See figure 1 herein and figure A-7 of MIL-PRF-28876. When mated to a plug or receptacle, the backshell shall hold the plug or receptacle insert in the proper position (see figures A-1, A-2, and A-6 of MIL-PRF-28876). Alternate dimensioning may be used for the backshell interface shown on figure 2 herein instead of the horizontal length dimensions shown in figure A-7, detail G of MIL-PRF-28876.

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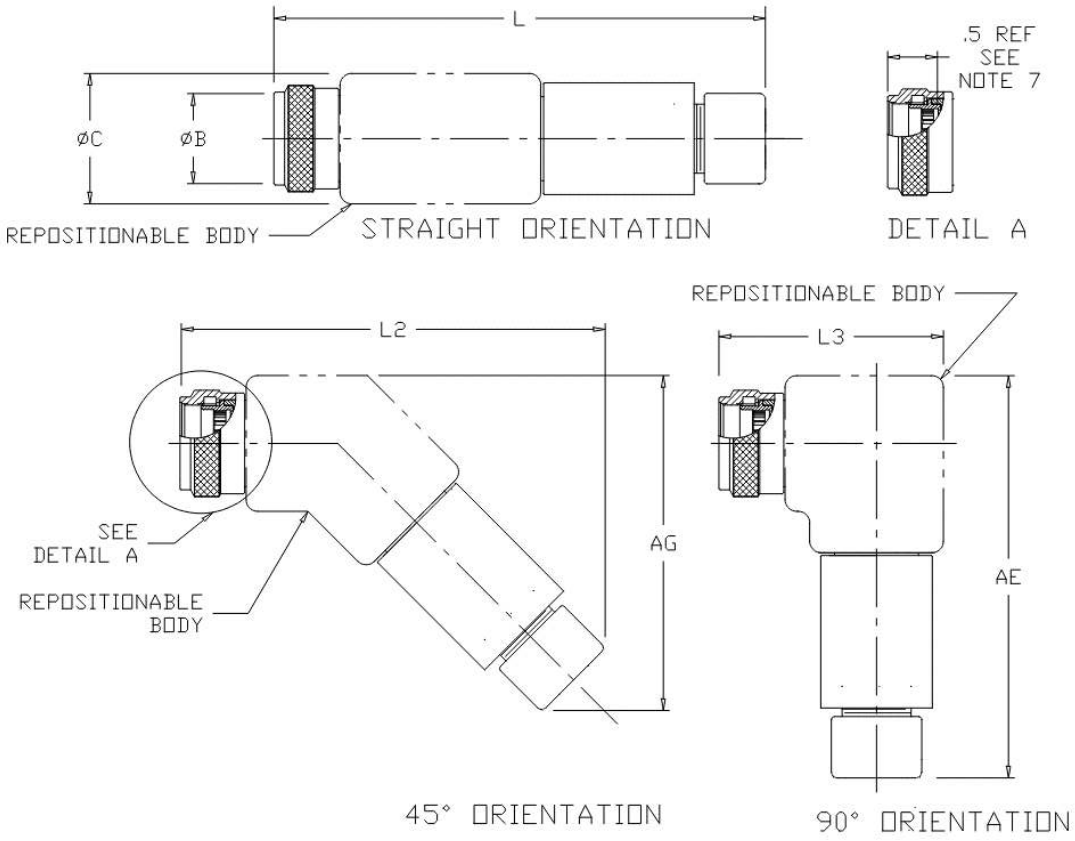


FIGURE 1. Repositionable backshell.

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Shell size	Shell size designator	Backshell designator	B dia max	C dia max	Straight	45-degree		90-degree	
					L max	L2 max	AG max	L3 max	AE max
11	A	1, 2, 3	1.06 (26.9)	1.43 (36.3)	5.1 (130)	4.5 (114)	3.6 (91)	2.4 (61)	4.3 (109)
13	B	1, 2	1.19 (30.2)	1.56 (39.6)	5.5 (140)	4.8 (122)	3.8 (97)	2.6 (66)	4.5 (114)
15	C	1, 2, 3, 4	1.31 (33.3)	1.75 (44.5)	5.9 (150)	5.1 (130)	4.0 (102)	2.8 (71)	4.8 (122)
23	F	1, 2, 3	1.75 (44.5)	2.13 (54.1)	5.1 (130)	4.4 (112)	3.9 (99)	2.8 (71)	4.5 (114)
		1L, 2L, 3L			7.4 (188)	6.7 (170)	5.1 (130)	3.8 (97)	5.8 (147)

NOTES:

1. Dimensions are in inches.
2. Metric equivalents (mm) are in parentheses and are given for general information only.
3. Dimensions apply to plated/finished part.
4. For mating spline (key) dimensions, see figure A-7 of MIL-PRF-28876.
5. Backshell internal configuration not shown. The backshell interface dimensions shall be in accordance with figure A-7 of MIL-PRF-28876.
6. Lengths L, L2, AG, L3, and AE are longer than found in other MIL-PRF-28876 backshell specification sheets (see intended use).
7. Distance from front of coupling nut to connector mating surface. This dimension is provided for reference only to determine horizontal dimension for a mated connector with backshell and is intended for installation design only.

FIGURE 1. Repositionable backshell – Continued.

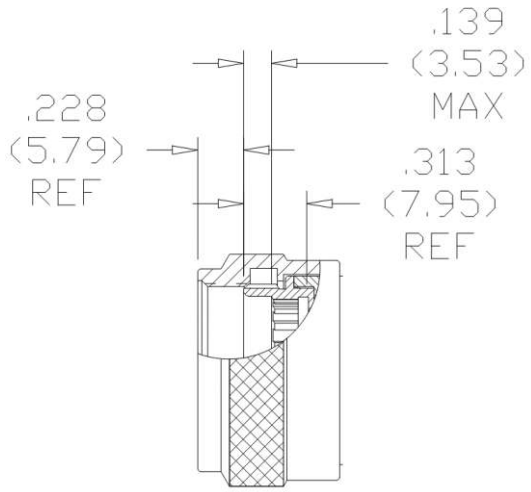


FIGURE 2. Alternate horizontal length dimensions for backshell interface.

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Weight.

Shell size 11: 7.5 ounces (212.6 grams), maximum.

Shell size 13: 9.0 ounces (255.2 grams), maximum.

Shell size 15: 11.5 ounces (326.0 grams), maximum.

Shell size 23: 22.7 ounces (643.5 grams), maximum.

Fiber optic cable.

Cable diameter. Backshells shall accommodate the cable diameters specified in table II herein.

Backshell coupling nut. Shall have spin coupling, meaning that the backshell coupling nut shall be captivated to, and free to rotate on, the follower of the adjacent circular portion of the backshell. This spin backshell coupling nut may be either non-self-locking or self-locking. The self-locking coupling devices may exhibit some mechanical resistance while captivated to the follower. When provided, the self-locking device shall be contained within the backshell coupling nut, shall be of a corrosion-resistant material, and shall provide a positive audible detented coupling (ratcheting). Connector-to-backshell couplings, either non-self-locking or self-locking, shall meet all the performance requirements specified herein. Backshells with self-locking coupling nuts shall meet the performance requirements without the use of a thread locking compound.

PERFORMANCE REQUIREMENTS

Mechanical properties.

Cable pull-out force. Shall be performed for repositionable backshells configured in the straight orientation.

External bending moment. Applicable. As specified in table I herein, shall be applied and held for 1 minute. Shall be performed on each test article at in the downward and sideward positions at load "P" (a total of two tests for each test article). The test articles shall be backshells with the straight orientation. Load shall be applied at a rate of approximately 10 inch-pounds (1.1 N-m) per second until the required load reached. The change in optical transmittance shall be measured after the test (see 4.7.3.4 of MIL-PRF-28876).

Load "P" shall be selected at a distance "L" from the rigid panel and located 90 degrees apart along a common circumference line on the backshell's exterior, in order to perform downward and sideward bending moments. Load "P" shall be determined as follows:

a. At a distance as far as possible from the face of the rigid panel, a solid, inflexible (i.e., non-elastomeric) section of the backshell body shall be selected.

b. The selected section on the backshell shall permit unimpeded access for the load inducing apparatus in the downward (vertical) and sideward (horizontal) radial positions. (NOTE: Alternatively, to avoid changing the radial position of the load inducing apparatus, the mated connector-backshell assembly can be revolved [i.e., clocked] 90 degrees on the rigid, vertical panel for performance of the sideward bending moment).

c. Any articulation features (e.g., pivot points, rotatable joints, etc.) shall be included within the distance "L" measured from the rigid panel face to the load application points.

Once the appropriate backshell section is selected, the distance "L", or lever arm, shall be measured (in inches) from the face of the rigid panel to the applied load "P". Refer to figure 2 of MIL-PRF-28876, which shows the backshell in a downward position. For a sideward position, reposition (rotate) the backshell 90 degrees along the backshell circumference. The distance "L" shall be the same for the applied load "P" in both downward and sideward positions since the two applied loads "P" lie along the same circumference line.

The load "P" is calculated as follows:

If bending moment (inch-pounds) = Length, "L" (inch) X Load, "P" (pounds), then:

Load, "P" (pounds) = Bending Moment (inch-pounds)/Length, "L" (inch)

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TABLE I. External bending moment.

Shell size letter designator (shell size)	Bending moment (inch-pounds)
A (11)	300
B (13)	300
C (15)	300
F (23)	520

Crush. Applicable. Shall be performed for repositionable backshells configured in the straight orientation.

Vibration. Applicable. The threaded articulation locking and accessory attachment elements of the repositionable backshells (see 3.14.15 of MIL-PRF-28876 and herein) shall not exhibit loosening or disengagement from their mated connections.

Backshell articulation. Ten repositioning cycles shall be performed by hand at a rate not to exceed one cycle per 5 seconds. A single cycle is the repositioning of the backshell from straight, to 45-degree, to 90-degree, to 45-degree, to straight. For each orientation during the 1st, 5th, and 10th cycle, the backshell shall be locked into place and tested for a positive lock, and an optical transmittance measurement shall be performed. The backshell shall show no evidence of mechanical defects or damage detrimental to operation. The change in optical transmittance shall be less than 0.5 dB during and after the test.

PIN, MATING PARTS, TOOLS

Part or Identifying Number (PIN). Marked on coupling nut of the backshell (see table II).

M	**28876**	**/32**	**-**	**X**	**X**	**X**
Prefix for military specification	Specification number	Spec sheet no.	-	Shell size designator (see table II)	Backshell designator (see table II)	Class designator (see table II)

Example: M28876/xx-B1

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TABLE II. PIN designators.

Shell size	Shell size designator	Backshell designator	Cable diameter range inch (mm)	Class designator (see table III)
11	A	1	0.240 - 0.250 (6.10 - 6.35)	T or blank
		2	0.280 - 0.315 (7.11 - 8.00)	T or blank
		3	0.305 - 0.346 (7.75 - 8.79)	T or blank
13	B	1	0.280 - 0.315 (7.11 - 8.00)	T or blank
		2	0.305 - 0.346 (7.75 - 8.79)	T or blank
15	C	1	0.495 - 0.515 (12.57 - 13.08)	T or blank
		2	0.380 - 0.423 (9.65 - 10.74)	T or blank
		3	0.423 - 0.465 (10.74 - 11.81)	T or blank
		4	0.305 - 0.346 (7.75 - 8.78)	T or blank
23	F	1L, 1	0.768 - 0.866 (19.50 - 22.00)	T or blank
		2L, 2	0.900 - 1.000 (22.86 - 25.40)	T or blank
		3L, 3	0.524 - 0.602 (13.30 - 15.30)	T or blank

TABLE III. Class designators.

Class designator	Description
Blank	Aluminum with non-conductive finish
T	Aluminum with conductive plating

Mating counterpart. Backshell mates with MIL-PRF-28876/1 receptacle, MIL-PRF-28876/6 plug, and MIL-PRF-28876/11 receptacle.

Installation and removal tools. Adjustable or 1.25-inch (31.8-mm) open-end wrench, strap wrench, and scissors. An additional tool may be permitted to loosen and tighten the adjustment for backshell orientation (articulation).

QUALIFICATION, CONFORMANCE, QUALIFICATION BY SIMILARITY

Qualification and conformance inspection. See table IV. This table specifies applicable requirements, qualification inspection, and conformance inspection per this specification sheet. The Requirement Paragraph and Test Paragraph columns within this table reference MIL-PRF-28876. A reduced sample set is allowed for conformance inspection; refer to MIL-PRF-28876 for conformance inspection sample size.

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TABLE IV. Qualification and conformance inspection.

Group	Qualification inspection tests [1/]	Backshell	Requirement paragraph	Test paragraph	Conformance inspection
I (six mated pairs)	Interoperability [2/]	X	3.4.3.2	4.8	--
	Visual and mechanical	--	3.10	4.7.2	--
	Size	X	3.10.1	4.7.2.1	A
	Weight	X	3.10.2	4.7.2.2	A
	Identification marking	X	3.10.3	4.7.2.3	A
	Workmanship	X	3.10.4	4.7.2.4	A
	Screw threads	X	3.10.5	4.7.2.5	A
	Functional [5/]	--	3.12	4.7.4	--
	Backshell and insert retention nut attachment	X	3.12.7	4.7.4.7	A
	Optical	--	3.11	4.7.3	--
	Insertion loss (initial)	X	3.11.1	4.7.3.1	C
	Return loss	X	3.11.5	4.7.3.5	C
II (two mated pairs)	Cable pull-out force	X	3.13.1	4.7.5.1	B [6/]
	External bending moment	X	3.13.2	4.7.5.2	B [6/]
	Cable seal flexing	X	3.13.3	4.7.5.3	--
	Twist	X	3.13.4	4.7.5.4	--
	Impact	X	3.13.6	4.7.5.6	--
	Crush	X	3.13.7	4.7.5.7	--
	Insertion loss verification	X	3.11.1	4.7.3.1	--
	Vibration	X	3.14.15	4.7.6.14	--
	Shock	X	3.14.16	4.7.6.15	C
	Water pressure	X	3.14.17	4.7.6.16	B [6/]
	Backshell articulation	X	3.1	M28876/32	C [7/]
III (two mated pairs)	Thermal shock	X	3.14.2	4.7.6.1	--
	Temperature/humidity cycling	X	3.14.3	4.7.6.2	C
	Temperature cycling	X	3.14.4	4.7.6.3	--
	Life aging	X	3.14.5	4.7.6.4	--
	Freezing water	X	3.14.6	4.7.6.2	--
	Sand and dust	X	3.14.7	4.7.6.6	--
	Identification marking	X	3.10.3	4.7.2.3	--

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TABLE IV. Qualification and conformance inspection – Continued.

Group	Qualification Inspection tests 1/	Backshell	Requirement paragraph	Test paragraph	Conformance inspection
IV (two mated pairs and parts)	Electromagnetic effects (two mated pairs 3/)	X	3.14.9	4.7.6.8	--
	Fluid immersion (two mated pairs)	X	3.14.10	4.7.6.9	--
	Salt spray (two mated pairs)	X	3.14.11	4.7.6.10	C
	Flammability (one mated pair)	X	3.14.12	4.7.6.11	--
	Fungus resistance (parts 4/)	X	3.14.13	4.7.6.12	--
	Ozone exposure (parts 4/)	X	3.14.14	4.7.6.13	--

NOTES:

1/ Group I mated pairs shall be used for groups II, III, and IV. Unless otherwise specified, the units undergoing testing shall be configured in the straight orientation throughout testing.

2/ Separate samples shall undergo interoperability testing. Connector interoperability samples shall be fully populated with termini on single fiber cable and configured without backshells. Backshell interoperability samples shall be constructed with terminated cables that fully populate the connector of the shell size under test.

3/ Perform only for connector samples utilizing non-metallic inserts.

4/ Polymeric parts from one mated pair.

5/ Six additional samples may be prepared for functional tests only.

6/ For group B testing, the following shall apply: one test sample - # 1, see table VI of MIL-PRF-28876, except retain in straight backshell orientation. Component-only test mated with connector and with cabling (see table VI, note 7 of MIL-PRF-28876), no optical measurements shall be performed.

7/ For group C testing, the following shall apply: two connector mated pairs - #1,2 (see table VI of MIL-PRF-28876).

Qualification by similarity for additional backshell sizes. If a backshell is qualified for a particular shell size, and a backshell of a different shell size with similar design, construction, and materials meets the visual and mechanical, size, weight, identification marking, workmanship, screw thread, backshell and insert retention nut attachment, external bending moment, vibration, and shock inspections herein, then the backshell of the second shell size is qualified.

NOTES:

Intended use. The repositionable backshell is intended for use where flexibility with the backshell orientation is needed. The repositionable backshell design and installation impact of a longer length must be considered over the advantage of more flexibility (repositioning) with the backshell orientation. Lengths L, L2, AG, L3, and AE on figure 1 herein are longer than found in other MIL-PRF-28876 backshell specification sheets.

Referenced documents. In addition to MIL-PRF-28876, this specification sheet references the following documents:

MIL-PRF-28876/1 MIL-PRF-28876/6 MIL-PRF-28876/11

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
DLA – CC

Review activities:
Army – AV, MI, TE
Navy – AS
Air Force – 11, 19, 93
MISC – MDA

Civil agencies:
GSA – FAS
NASA – NA

Preparing activity:
Navy − SH
(Project 6060-2023-020)

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